Harshita Sharma

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EDUCATION

B.Tech, Mechanical Engineering — IIT Bhilai | CGPA: **8.29**/10 | Best SGPA: **8.6** 2024 – **2028**

EXPERIENCE

Machine Learning Engineer Intern — APTransco, State Load Dispatch Centre *May – July 2026*

- Diagnosed a production forecasting failure caused by stale SCADA telemetry and engineered staleness-aware features so the model degrades gracefully instead of failing silently.
- Designed a staleness-gated **ensemble** combining XGBoost, LightGBM, and a **Transformer** sequence model, validated with causal walk-forward recalibration over **120,808** leakage-free blocks.
- Reached **1.66% MAPE** next-block (**3.91%** under a 24-hour blackout); shipped as a live Linux service with a JSON API refreshing every 60s.

Tools: Python, PostgreSQL, XGBoost, LightGBM, PyTorch (Transformers, Attention), Scikit-learn, Pandas, NumPy, Linux, Git

PROJECTS

Autonomous Precision Landing of Quadrotor — ROS 2 / PX4 Repo

- Built a multi-process ROS 2 (Humble) system with OpenCV ArUco detection, a finite-state landing controller, and two PID loops for PX4 offboard control in Gazebo SITL, each process independently testable.
- Integrated PX4 and ROS 2 through the uXRCE-DDS bridge, exposed controller gains as runtime parameters, and implemented marker-loss recovery plus a controlled final descent.

Tools: Python, ROS 2 Humble, PX4 SITL, uXRCE-DDS, OpenCV, Gazebo, Linux, Git

Lexa — Indian Languages Learning App Repo

- Building a full-stack web app (Telugu, Tamil, Kannada, Hindi) with interactive lessons, progress tracking, authentication, and a responsive UI.
- Built reusable React/Next.js components on server-side API routes connected to PostgreSQL via Drizzle ORM, with automated lint/build testing and database seeding.

Tools: Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, Tailwind CSS, Clerk, Git

Asteroid Risk Classification — Intra-IIT ML Challenge (1st Place) Repo

- Built an end-to-end pipeline covering data cleaning, feature engineering, and class-imbalance handling on a heavily imbalanced dataset; won **1st place**.
- Traded overall accuracy (**82%**) against hazardous-class recall to match an asymmetric-cost decision.

Tools: Python, Pandas, NumPy, Scikit-learn, Random Forest, XGBoost, Jupyter

Grant Tracker — Blockchain Grant Management Repo

- Built a React frontend for transparent grant submission, review, approval, and disbursement, backed by a Soroban (Rust) smart-contract service on the Stellar Testnet.
- Implemented applicant/admin workflows, wallet authentication, and on-chain state transitions with lifecycle guards against invalid operations.

Tools: React, JavaScript, Rust, Soroban, Stellar SDK, Freighter Wallet, Tailwind CSS, Git

Other Projects

Traffic Demand Prediction-(Gridlock Hackathon Flipkart) — stacked a LightGBM/XGBoost/CatBoost ensemble with a multi-person team to forecast city-wide demand, reaching OOF R^2 **95.43** under leakage-free CV. [Notebook]

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, **Transformers & Attention Mechanisms**, Sequence Modeling, Scikit-learn, XGBoost, LightGBM, CatBoost, Time-Series Forecasting, Feature Engineering, Ensemble Methods, Pandas, NumPy
Web & Data: PostgreSQL, Drizzle ORM, Next.js/React, Tailwind CSS
Languages: Python, C++, C, Rust, TypeScript, Bash, SQL
Software Engineering: Git/GitHub, **Pull Requests & Code Review**, **Automated Testing & Test Coverage**, Debugging & Root-Cause Analysis, CI/CD, SDLC, Agile/Scrum, Navigating Unfamiliar/Multi-Language Codebases, Linux
CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis, Relational Databases, Dynamic Programming, Problem Solving

ACHIEVEMENTS AND LEADERSHIP

- **Maintainer, OpenLake (IIT Bhilai)** — own and help run a **multi-contributor open-source project**: review **pull requests** for code quality and **test coverage**, give and receive **technical feedback** in design discussions, triage issues, and coordinate distributed contributors via Git.
- **Competitive Programming — DSA in C++/C:** Codeforces (**120** problems), LeetCode (**50-Day Badge 2026**; 75+ solved in **C++**).
- **Core Member, Blockchain Society (IIT Bhilai)** — organize technical workshops and hackathons end-to-end.
- **3rd Position, Summer ChainRxn Bootcamp** — developed an animated **front-end landing page** applying modern **UI/UX** principles [Repo 1] [Repo 2].
- **MATLAB Certifications:** ML Onramp; Classification Models [Certs]